Andrew Olvera

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RESEARCH INTERESTS

Agile, high-speed robotics: learning representations and control policies that generalize across environments. World models, motion planning, perception, and SLAM for robust autonomy in complex and GNSS-denied settings.

EDUCATION

York College of Pennsylvania | Aug 2023 – May 2026

B.S. in Computer Science, York, PA

Selected coursework: Linear Algebra, Probability & Statistics, Computer Vision (capstone project), Algorithms, Computer Graphics

Cabrini University | Aug 2021 – May 2023

Undergraduate coursework, Radnor, PA; transferred to York College

RESEARCH EXPERIENCE

Starscream | 2026 – Present

Independent research | *Agile drone racing* | *Manuscript in preparation*

- Developed a hierarchical racing policy that plans motion through a continuous 64D latent interface; a pretrained, frozen feedback controller maps the planned motion and current state to actions.
- Built a staged pretraining pipeline: action-conditioned world-model backbone; expert-motion encoder and PPO-trained controller; JEPA-style latent alignment; and physical alignment through an analytic differentiable plant before planner RL.
- Trained the planner with racing PPO, dynamic course sampling, and online course generation.
- The evaluated 349K-parameter actor operates at 130 Hz in nominal simulation with perfect state and gate-map inputs.
- Designed geometric evaluation bands to measure performance as courses depart from the training distribution.
- A single-seed development evaluation achieved 85.8% clean completion in-distribution (309/360), 38.3% in the near band (69/180), and 32.2% in the outer band (58/180).
- Compared against a scratch direct-action MLP: the selected full-system checkpoint improved near/outer-band completion by 19.4/20.6 percentage points. Analyzed matched-success lap times and architecture/pretraining variants.

Tells | 2026 – Present

Independent research | Opponent modeling and model-based RL | Manuscript in preparation

- Built MicroRTS state/action tokenizers and action-conditioned latent world models for complete-state and fog-visible representations.
- Investigating 1M- and 5M-parameter dynamics models and reward/termination prediction.
- Distilled a 16-code opponent belief from fog-of-war history using privileged training targets. Studied return prediction and belief-conditioned policies, with downstream policy benefits still under evaluation.
- Implemented PMPO actor-critic learning with eight-step imagined rollouts in a frozen learned world alongside real-engine data collection; developing controlled opponent-latent and policy-learning ablations.

Reward Design in Small-Model RLVR | 2025 – 2026

Independent research | Site report

- Studied Qwen3 0.6B on GSM8K, comparing GRPO and DAPO group sizes and binary, structured, and rank-based rewards.
- Exploratory single-seed runs reached approximately 69–72% accuracy from a 68% SFT baseline.
- Analyzed token-level advantages, dynamic clipping, and reward variation to understand learning signals as binary rewards saturated.

INDUSTRY EXPERIENCE

Sapio Sciences | May 2025 – Present

Application Engineer Intern | York, PA

- Build LIMS/ELN data models, workflows, and integrations on Sapio's lab informatics platform, translating biotech and pharma research requirements into deployed solutions.
- Contribute platform infrastructure changes and root-cause debugging across support, product, and engineering teams.

Dataforma | May 2025 – Jan 2026

Software Engineer Intern, Web Development | York, PA

- Refactored legacy JSP features onto a Vue.js frontend and Kotlin backend; collaborated with development, design, testing, and QA teams to implement features and validate releases.

TECHNICAL SKILLS

ML & RL: PyTorch, Hugging Face Transformers, TRL, vLLM; imitation learning, policy gradients, model-based RL

Robotics: MPC/MPCC (acados), motion planning; ROS, Flightmare, Isaac Sim, MATLAB/Simulink

Programming & tools: Python, C++, C, Java, Kotlin, JavaScript, SQL; Linux, Git, Docker, Bash, Weights & Biases, FreeCAD

HONORS

Dean's List, York College of Pennsylvania | Fall 2025, Spring 2026